

Review

Manual Catching and Transportation of Poultry with a Focus on Chickens and European Practices

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Abstract

The manual handling of poultry is an essential part of raising and caring for birds. The different manual catching methods have various impacts on the bird's welfare and health, the well-being and work satisfaction of the people who are handling the birds, and the economic and logistical requirements of everyone involved. The traditional approach of using the Five Freedoms for investigating animal well-being has been amended with animal-based measures (ABMs) as well as the evaluation of five welfare domains, which consider the subjective response of an animal towards its environment. The assessment of single individual animal welfare parameters without context can be non-specific, only partially informative, or even misleading when considered in isolation. The objective measurement of suitable parameters for the evaluation of the various steps of poultry catching and transport is complex and should be carried out in a differentiated manner. This review summarizes the current knowledge about the manual catching of poultry, with special focus on the upright and inverted handling of chicken and current considerations in Europe. The implementation of consistent, transparent, and traceable central data collection on animal health and welfare at various critical control points of bird transportation would allow systematic evaluation of the multifactorial welfare assessment in the future.

Keywords: bird; depopulation; hen; broiler; handling; health; welfare; stress

1. Introduction

There are various occasions that require manual catching and transportation of individual birds or an entire flock. Whether it is during random sampling of individual birds during daily routine checks, for the welfare assessment of weekly evaluations (such as assessing body weight, feather cover, or foot health), when capturing, examining, and transporting sick or injured animals, for individual bird treatment such as intramuscular injections for vaccination, or for the purpose of capturing and loading for transport to another facility or for slaughter. While the legal regulations of handling and transporting poultry differ globally from no regulations to detailed national laws, it is worldwide common practice to catch birds by one or two legs and carry one or several of them in inverted position for a short distance to their destination (e.g., the scale, the place of medical observation or to the vaccination machine, into the transportation crate) [1–5].

The focus on bird welfare in the European Union (EU) during transport is regulated in detail by applying the following provisions:

- “No one may cause pain, suffering, or harm to an animal without reasonable cause.” ([6] §1).



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- “No person shall transport animals or cause animals to be transported in a way likely to cause injury or undue suffering to them.” ([7] Article 3).
- “The personnel handling animals are trained or competent as appropriate for this purpose and carry out their tasks without using violence or any method likely to cause unnecessary fear, injury or suffering;” ([7] Article 3 (e)).
- “It shall be prohibited to: [...] b) apply pressure to any particularly sensitive part of the body in such a way as to cause them unnecessary pain or suffering; [...] d) lift or drag the animals by head, ears, horns, legs, tail or fleece, or handle them in such a way as to cause them unnecessary pain or suffering;” ([7] Annex I Chapter III No. 1.8 b & d).
- “[...] However, the prohibition on lifting animals by their legs shall not apply to poultry, rabbits and hares;” ([8] Annex III No. 1.8 c).

The regular revision of these laws and regulations is usually accompanied by input from various interest groups, which seek to reconcile potential contradictory aspects addressing animal welfare, societal acceptance, logistic feasibility, meat product quality, and economic cost-benefit analyses [9,10].

The aim of this review is to present current scientific knowledge on the manual catching and manual handling of poultry with special focus on chicken, thus contributing a factual, evidence-based input to the aforementioned debates. Due to the holistic approach of this review, which covers relevant anatomical, physiological, individual-based, flock-based, and logistical/economic information, the focus of this review was limited to manual catching. Readers interested in mechanical catching technologies as used in broilers are referred to other literature such as Werner et al. (2023) [11].

2. Literature Search and Study Selection

A scientific literature search on the required topic using the keywords “poultry” and/or “transport” and/or “catching” and/or “carrying” and/or “chicken” and/or “bird” and/or “hen” and/or “upright” and/or “position”, and/or “welfare” and/or “stress” and/or “handling” was conducted using common search engines (Google Scholar, PubMed, Research Gate, Scopus) with no restriction to publication date. The received references were screened and removed in case of duplications, non-English and non-German language or including information on non-target species. Publications with high standards considering evidence-based quality evaluation were limited: No meta-analyses or systematic reviews could be identified and besides randomized controlled trials and field studies with variable controllable study conditions, textbook knowledge, individual studies, and observations, some of older dates, were also used for this review. Where necessary, existing knowledge of other poultry species (ducks, birds of prey) was included. Therefore, the following were used in this report: international and national scientific, peer-reviewed primary literature, dissertation theses, summaries and reports from governmental institutions and committees, works created on behalf of non-commercial institutions such as research communities, conference reports and textbook content. A selection bias may remain given this narrative approach.

3. The Evaluation of Animal Welfare in Poultry

While the welfare of an animal has traditionally been defined by the concept of Five Freedoms (freedom from hunger, thirst and malnutrition; discomfort; pain, injury and disease; fear and distress; freedom to express normal behavior), there is increased establishment of positive animal welfare parameters and the assessment of behavioral patterns to capture the status of well-being [12–16]. However, due to the limited accessibility of behavior and housing of poultry during the restriction by manual handling and transport, both

negative behaviors (e.g., avoidance behavior, withdrawal, freezing, stereotypies like head shaking, feather pecking, circling movements) and positive movement patterns such as exploratory behavior or successful coping strategies are only partially obtainable [17–21]. In addition, the evaluation of subjective animal-based parameters characteristic for a positive emotional state such as satisfaction, playfulness and comfort cannot be reliably evaluated during stressful events in chicken flocks [15]. Despite these limitations, parameters listed in Table 1 have been investigated to evaluate the well-being of birds and could be used for their evaluation during handling and transportation:

Table 1. Parameters to evaluate the well-being of birds. These parameters are not specific and some may be limited to capture the body’s short- or long-term response. With many of the parameters being modified by multifactorial influences, careful data interpretation considering the experimental conditions must be applied. The list is not exclusive.

Parameter	Individual-Based	Flock-Based
Behavior	Restlessness, vocalizations, feather pecking, escape attempts, frequency of feces passage, chicken stress scale, tonic immobility test, novel object test, avoidance distance test, avoidance distance touch test, approach test	Smothering events, sound volume & vocalizations, movement flow
Physiology	Body temperature (cloacal & comb temperature), respiratory rate, beak breathing, heart rate/pulse	
Biology	Hematocrit, creatine kinase, fat metabolism products, adrenal gland weight, antibody formation/immunological parameters, hormones (e.g., corticosteroids, prostaglandins, adrenocorticotrophic hormone (ACTH)-stimulation test)	Hormones (particularly corticosteroids)
Health	Physical integrity of the living animal or carcass (condition of feathers, skin, body structure, organs, tissues including bruises, fractures, epiphyseal separations), necropsy findings	Flock treatment rate, mortality rate, rate of death on arrivals (DOA)
Performance	Body weight (change), ovary activity upon necropsy	Feed intake, body weight change, laying rate

While one individual parameter alone has limited value and can even be misleading; an integrated and holistic assessment of various parameters on multiple levels is known to reflect, more accurately, the actual biological condition of the animal [22–25]. In particular, the evaluation of stress is considered substantial during handling/transport, and many attempts were made to quantify its burden to the animal [26–29]. Stressors are events that pose a real or perceived threat to homeostasis. The body responds to these events with a biological reaction, which can be physiological, immunological, cognitive, or emotional in nature, and is usually also reflected in the behavior of the animal [30]. Negative distress (in contrast to positive eustress) manifests itself in that the biological compensation mechanism is insufficient, and an adequate physiological response is absent [31].

The definition of “handling stress” was mentioned in a statement by the European Food Safety Authority (EFSA) on the transportation of animals as one of 33 potentially affected animal welfare consequences and is considered a result of the interaction with human or machine influence [32]. Since the interaction between humans and animals during capture always involves an (unusual) contact and restriction of movement for the animal, it can be assumed that chickens experience not only stress but also other negative emotions during capture, regardless of the method such as [33]:

- Fear: an unpleasant negative situation induced by the perception of a (potential) danger that could threaten the integrity of the animal [34],
- Frustration: a negative emotional evaluation of the situation as consequence of being unable to satisfy a need. This is often triggered by a hindrance to the exercise of natural behaviors.

Being exposed to various stressor types during transport, the sum of these stressors has an additive effect on the animal. Individual stressors include, for example, the altered microclimate of the new environment, the acceleration and braking forces of the transportation vehicle as well as its vibrations, the animals' food and water deprivation, social stress, and unfamiliar noises including loud sounds [23,35,36]. These factors can lead to a wide range of impacts on the individual animal, ranging from the feeling of mild discomfort to anxiety and pain, even resulting in death [37,38]. In order to evaluate an individual stressor during a certain part of the transportation event (Figure 1), for example, to isolate the influence of the method of manual catching and manual carrying, a temporally coordinated and precisely defined objective evaluation of a representative number of individual animals is necessary, as well as a method and execution that is sensitive enough to distinguish the effects of the different catching methods from each other and from preceding or subsequent stressors.

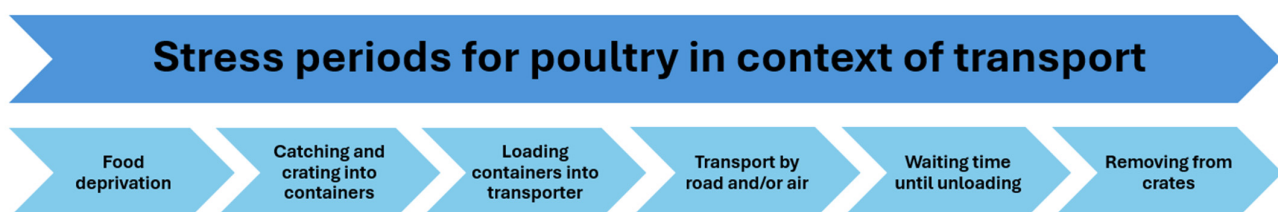


Figure 1. Stress periods for poultry in context of transport, which can affect animal welfare.

For example, in broilers that were placed in transport containers after being caught, the stress caused by the type of handling could not be measured using blood corticosteroid concentration, as being confined in the transport containers itself represented a far greater stressor than the handling event, hence overriding the corticosteroid response of the handling that occurred before the crating [39]. The challenge of finding a suitable method to specifically evaluate the influence of only *one* process (e.g., only the catching, not the carrying or crating) within a chain of events needs to be considered in the interpretation of the data [40,41]. Similarly, when evaluating commercially caught chickens, the restrictive environment in the transport crate masks both the influence of different handling methods and the stress caused by food deprivation [39,42,43]. Similar observations were made by Beuving and Vonder who observed a continuous increase in corticosteroid concentration in caged laying hens up to four hours after being placed in transport crates, which reliably led to non-interpretative corticosteroid values of the catching process when the animals were placed in transport containers prior to blood sampling [44,45]. Cold and heat stress, handling, and the resting state of the flock also influence the corticosteroid levels in chickens and should, therefore, always be recorded as a baseline and in a control group [46,47].

Additionally, genetics, diurnal hormonal fluctuations, and the effects of short-term and long-term stress must be taken into account, as the body makes biological long-term adaptations (e.g., to housing conditions) that change the biological response [28,48]. The use of blood concentrations of various other hormones, enzymes, or metabolites such as creatine kinase or glucose can also serve as indicators of stress levels but are multifactorially influenced by the animal's age or the condition of its musculature [29,42,43,49,50]. The increased gluconeogenesis in stressful situations also results in the breakdown of body proteins and subsequently can lead to an increase in the total protein concentration in

plasma, which again is not specific [43]. Similarly, the extent of weight loss can be used as an indicator of the well-being of poultry, but it is also indicative of an overall situation (e.g., the intensity and duration of the animal transport as a whole) and less for capturing the individual steps of the specific event (catching, holding, carrying, placing into the transport container, loading, transporting, unloading) [42,51,52]. Overall, the type, severity, and duration of the relevant impairment of the additive stressors to animal welfare should always be considered. Regarding this assessment, this means considering not only the type of manual catching and carrying, but also the duration of the method and the severity of the consequences, especially the effects on other individuals of the flock of concern.

4. Comparative Anatomical–Physiological Features to Be Considered for the Individual Animal

4.1. Carrying Chickens in an Upright Position Is Commonly Performed When Handling Small Numbers of Individual Birds

Holding in an upright position can be defined as the upright catching and holding of poultry in a person's hands (Figure 2). It is important that the pressure applied to the animal is just strong enough to avoid its escape, but not so strong that it causes fear and pain [53,54].



Figure 2. Upright handling of domestic chicken. The chest is supported while the hands cover the wings, preventing flapping. Two birds next to each other also hinder wing flapping. The Figure was provided from Prof. Dr. Ruhnke's private photo archive with permission for publication.

In an upright position, the internal organs remain in a physiological body posture (Figure 3). The coelomic cavity of birds is NOT divided by a muscular diaphragm like in mammals, but only by some membranes [55]. The lungs are firmly attached dorsally, meaning they are adjacent to the spine, and, in contrast to mammalian lungs, are a rigid structure expanding only slightly (about 1%) during exhalation [56,57]. The ventilation of the lungs occurs with the help of nine air sacs in most domesticated poultry (one clavicular sac, two cervical sacs, two caudal thoracic sacs, two cranial thoracic sacs, and two abdominal sacs), which allow the avian lungs to be supplied with air during both inhalation and exhalation [58–62]. The air sacs extend into the interior of various bones, such as the femur, the humerus, the thoracic vertebrae, some ribs, the sternum, and the pelvis, forming a continuous volume reservoir [61,63]. Which and to what extent the aforementioned bones are equipped with air sacs greatly depends on the species of bird,

the humerus and the base of the sternum are the dominant air-filled bone structures in chicken [64]. Birds achieve an efficient oxygen yield through this lung-air sac system. The volume of the air sacs can adapt to the bird's position, the filling state of the digestive tract, and the reproductive cycle of the animal, so that despite space-consuming structures in various places (e.g., the presence of eggs), sufficient oxygen can be exchanged [64,65].

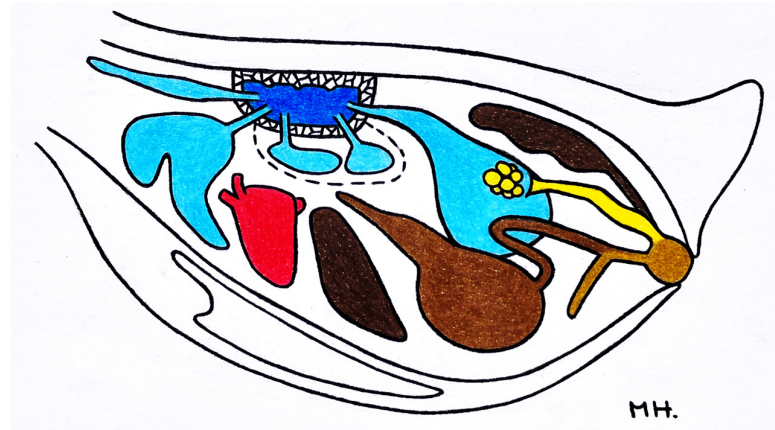


Figure 3. Representation of the coelomic cavity of birds using the domestic chicken as example, with their air sacs in relative position to other organs in an upright posture (Adapted with permission from Ref. [66]. 2009 Schattauer GmbH.). The respiratory organs (lungs (dark blue) and air sacs (light blue)) are depicted in blue, while the other organs such as heart, liver, parts of the stomach/intestines, kidneys, oviducts, and cloaca are shown in red, brown and yellow, respectively. A distinctive feature of birds are the lungs, which are rigidly fixed to the vertebrae and dorsal part of the ribs, being passively ventilated during both inhalation and exhalation.

The distribution of airflow in the bird's respiratory system depends on the strength and intensity of the respiratory muscles, as well as the mechanical interplay of the body wall, lungs, and air sacs. The airflow is generated by the active movement of the ribcage during inhalation and exhalation (Figure 4). Both inspiration and expiration in birds are only possible through active muscle contractions, through which the bone movements either expand or reduce the volume of the air sacs, thus supplying air to and from the lungs during breathing. Breathing due to active diaphragm movements does not occur in poultry [60].



Figure 4. Changes in the thoracic skeleton during the breathing process in birds (frontal view). The solid black line represents the position of the ribcage at the end of expiration. The red drawing shows the position of the ribcage at the end of inspiration (Adapted with permission from Ref. [67]. 2008 Blackwell Publishing.).

During inspiration, the sternum moves cranially and ventrally, while the coracoid and the furcula move cranially [68]. The ribs are also lifted cranially, thereby expanding the coelomic cavity in circumference [67]. It is therefore essential that the mobility of the

sternum is restricted as little as possible by manually holding the bird, as otherwise this could lead to suffocating the bird [65,68,69]. De Lima et al. noted that when broilers were caught and carried upright individually, they exhibited less agitation in the catcher's hand (OR = 0.25) and in the cage (OR = 0.57) and expressed less defensive movement during transfer to the cage (OR = 0.22), compared to broilers that were carried upright together side by side in one hand [2]. This could possibly be a result of the less restricted mobility of the ribcage during individual transport.

The term “inverse holding of poultry” refers to the short-term overhead holding of birds, where the animal is carried by one or two legs (Figure 5). The leg is grasped at the metatarsophalangeal joint, below or at/above the ankle joint [70]. One or more animals can be carried by hand.



Figure 5. Short-term inverse position of a domestic chicken (Adapted with permission from Ref. [66]. 2009 Schattauer GmbH.).

The ventilation of the lungs has been studied in various bird species; however, most studies focus on the locations of oxygen receptors or the respiratory behavior of raptors, pigeons, ducks, and chickens, which lie on their breast or back [71,72]. Early studies from the 1960s concluded that the air sacs can maintain a constant ventilation pressure in the lungs due to their variable size [73]. During supine positioning, which is regularly performed during medical care under anesthesia, the breathing frequency and oxygen saturation may be maintained, even though the internal organs get displaced in such a way that the breath amplitude in dove, duck, and chicken is significantly reduced [72–75]. In the Red-tailed Hawk, it has been shown that despite impaired air sac ventilation and increased tidal volume in the supine position, the oxygen saturation of the animals was not adversely affected due to adaptation of the parabronchial ventilation [76]. It is believed that in the supine position, especially, the liver and the reproductive tract of the female hen partially displace the caudal air sacs, but the lung itself, due to its location and protection by connective tissue aponeuroses, is not compressed or otherwise impaired even in obese animals [64,72,77–79]. For this reason, but also based on clinical experience, not only birds of prey but also pet birds, waterfowl, and poultry are regularly placed on their backs during anesthesia [75,79–82]. However, data collected in supine position under anesthesia cannot be directly applied to the short-term inverted position of poultry.

4.2. *The Conditions and Duration of the Upright and Inverse Capture Methods and Carrying Position in Birds*

The upright body position is generally used by all birds either standing, sitting, or lying down. In addition, the inverted position is physiologically adopted briefly by raptors, cormorants, kingfishers, and other birds of prey during a dive, as well as by aquatic birds and waterfowl during dabbling and diving processes. Except for the emperor penguin, birds voluntarily do not dive for longer than 1–2 min, and most birds dive for no longer than one minute [83]. For ducks, a change in the position of the head in diving position alone does not cause respiratory arrest; it is only when the head is submerged under water that a pause in breathing occurs [84]. In the case of turkeys, broilers, young hens and laying hens, manual catching and transportation occurs commonly in an upright position or by the legs, followed by lifting and placement in the inverse position for a short period of time [85–89].

Individual turkeys, ducks, and geese are caught upright for individual assessments, while for loading, several turkeys, ducks, and geese are herded in small groups with a defined maximum number of animals close to the transport containers and then manually caught upright and transferred [90–94]. Alternatively, the animals are also guarded directly into the transport containers, allowing them to walk in without direct human contact [90–94]. Ducks and geese must not be caught by the legs or transported invertedly [90,91,94]. ‘Heavy’ ducks and any goose must be caught individually and carried upright. There is limited scientific evidence and, to our best knowledge, no studies of ducks and geese caught and carried by their legs have been published. The legs of ducks and geese are, in relation to their body weight, relatively shorter and weaker compared to chickens. Hence, catching ducks by their legs is not feasible. Carrying ducks and geese by the legs would provide a high-risk activity for fractures and is often explicitly prohibited. Traditionally, ducks and geese are grasped by their necks, allowing them to be caught as calmly and quickly as possible to ensure smooth transport. The weight of all Musk and Peking ducks must be supported by one hand under their body and one arm around their body, holding the wings in a closed position. In the EU, unnecessary carrying and holding of commercial birds is clearly prohibited [91–94]. Geese are lifted with both hands around the core of the body or at the base of both wings, with one arm placed around the body for carrying and the wings held in a closed position [95]. The other hand supports the upper neck to prevent the goose from biting (occupational safety) [94]. Exact time specifications are not provided, but it is noted that the transfer to the transport containers must take place “without delay” [94].

Restrictions on animal movement using an inverse position are also applied when slaughtering individual birds (mostly in farm slaughtering of geese, turkeys, or ducks) by placing them in a slaughter funnel or for animals weighing under 3 kg live weight during manual neck breaking [54,96]. For catching, transporting and crating pullets to their laying shed, slow-growing broilers to their fattening shed, and the transport of broilers and laying hens to the slaughterhouse, individual animals are usually constrained for about 8–10 s until they are placed in the transportation containers laced in close proximity. In contrast, hanging broilers and laying hens by two legs in the slaughter hooks of the conveyor chain from the transport cage to the stunning in the electric bath is permissible for up to 60 s ([8]—Annex II, 5.2.).

The manual catching and conveyance of poultry in Europe is carried out by professional catching teams and is done here, as well as worldwide, almost always by the legs and inverse [37,97,98]. One exception is the Netherlands where inverted catching and carrying is prohibited due to a court ruling from 2022 [99]. Other countries like Switzerland lack skilled labor to provide professional catchers [100]. Non-caged pullets and laying hens are always caught manually, as their structured housing system over several aviary levels

makes the use of mechanical catching machines impossible with the current systems. The Animal Welfare Committee of Great Britain recommended the maximum manual handling duration of 5 s to 2 min for such birds and a distance between the catching and loading location of the poultry to be no further than 50 m [69]. In contrast, in Germany, the loading containers are usually taken directly into the shed, so the transport distance for the poultry is a maximum of 1–3 m and the carrying duration is generally a few seconds. Exemptions may occur in mobile sheds where transportation containers may not fit into the hen house.

During inverse carrying, the legs of the carrying human can support the weight on the animals' chests; no swinging or rotating of the animals should occur [54]. To prevent hip dislocations and other injuries, carrying broilers by two legs has been recommended [101,102]. In some countries, including Norway, Finland, and Germany, broilers are also caught and loaded upright using machinery, which, when using the appropriate equipment and trained personnel, results in comparable welfare results [3,97,103,104].

5. The Influence of Catching and Carrying Poultry by One and Two Legs, as Well as Holding and Transporting Poultry in an Upright and Overhead Position

The influence of catching and carrying broilers by one or two legs was very thoroughly investigated by Wessel et al. [70]. Twelve transports from seven commercial broiler farms with holding capacities of 20,000 to 50,000 broilers were documented using video cameras and evaluated regarding their behavior and injuries [70]. The animals were between 30 and 40 days of age and weighed between 2.3 and 3.1 kg. At each farm, the broilers were randomly caught by one or two legs and carried invertedly to be placed in the transport container. Various influencing factors (e.g., the grip at different leg joints such as the metatarsal joint, the hock joint, or above the hock joint) as well as carrying a bird without or with up to four additional chickens were investigated. Furthermore, the broilers were examined the day before transport to take any pre-existing conditions into account. Behavior tests were conducted to quantify the fear response to interaction with humans (inversion itself can cause prolonged tonic immobility, which can be interpreted as a fear reaction [105]). In total, 1.1% of the broilers that were caught and carried by one leg, as well as 0.43% of the broilers caught by two legs, experienced epiphysiolysis as a direct consequence of handling. The probability of epiphysiolysis was significantly higher in broilers caught by one leg ($p = 0.042$) [70]. Placing broilers in the lower transport containers and catching them in the fifth/last working hour had a significant negative impact on the incidence of epiphysiolyses, and female broilers were affected significantly more often [70]. Catching and carrying the broilers by one or two legs had no significant effect on the prevalence of bruises, which is in line with Langkabel et al. [106]. Animals that were placed in the lower transport containers, as well as those that flapped their wings more frequently, bruised more frequently [70,106]. However, the number of wing flaps during the catching and while sitting in the transport boxes did not differ between broilers caught by one leg and two legs [70]. During the carrying process itself, however, broilers that were caught by one leg and carried invertedly flapped their wings significantly more often (88.0%) compared to broilers that were caught by both legs and carried invertedly (80.6%) [70]. Other factors contributed significantly to the reduction in wing flapping: Grasping around or above the ankle as well as carrying multiple animals at the same time significantly reduced wing flapping. Hence, when carrying several animals simultaneously, there was no statistical difference between the injuries of the one-legged and two-legged caught broilers [70]. It demonstrated that it was not the carrying by one or two legs that was relevant for animal welfare, but rather the accompanying circumstances and the manner in which catching and carrying was conducted. These observations are consistent with

the study by Wolff et al., who evaluated 17 commercial broiler farms and found that broilers remaining in physical proximity to each other behaved more calmly and exhibited fewer injuries [103]. In contrast, broilers housed and caught in research facilities which were carried invertedly in groups had increased corticosteroid concentrations compared to broilers that were either individually caught by the legs and carried overhead or broilers that were held upright [39]. These broilers were housed in a scientific facility with nine animals per group, so it can be assumed that the handling and holding of the animals were recorded very precisely. The catching process, however, unlike in a commercial barn, allowed fewer escape options; therefore, the stress response and duration of the entire catching process can only be partially transferred to the situation of broilers in large flocks [39].

Langkabel et al. investigated the impact of one or two leg catching and concluded that the catching method had no effect on the prevalence of bruises or fractures in the carcass [106]. However, broilers caught by two legs made more escape attempts during the catching process, were more restless, and flapped their wings more often [106]. The difficulties associated with catching by both legs may have contributed to the animals acting more stressed and restlessly, thus increasing their risk of injury [106]. In Kittelsen et al., two-legged inversely captured broilers were more restless than upright captured animals [107]. This resulted in wing fractures for seven of the 2010 two-legged captured broilers, while only one of the 1941 upright captured broilers was injured in this way. Delanglez et al. found that broilers showed fewer wing flaps during the upright catching method (2.0 ± 0.1 vs. 5.4 ± 0.1 on a 7-point Likert scale, $p < 0.001$) and had better catcher–bird interaction (3.7 ± 0.2 vs. 4.4 ± 0.2 on a 7-point Likert scale, $p < 0.01$) compared to the inverted method [3]. However, the upright method exhibited a significantly higher incorrect application of the catching method ($22.43 \pm 12.3\%$ upright vs. $0.41 \pm 0.60\%$ inverted, $p < 0.001$) [3].

The issue of animal collisions can be particularly observed during the upright handling of relatively agile laying hens. Gerpe et al. observed the active (directly caused by the handlers) and passive (caused by the hens' movements) collisions with the housing equipment [100]. The study found that 8.1% of layers suffered severe injuries (fractures, muscle damage, and/or elevated corticosterone levels) due to collisions during the catching which was carried out by non-professional catchers (including family members and neighbors). Half of the collisions were caused by the animals themselves [100]. While in floor-reared broiler housing, feed chains and water pipes can be raised to the ceiling and in such a way removed before catching, this is not possible with the tiers in the vertically structured aviary housing of pullet and laying hens. Therefore, the risk of collisions is higher in non-caged layers. A successful, quick, and calm capture is therefore of most importance for a successful first catching attempt of the layers. This is particularly important for hens strains that react faster and act more flighty. According to a study by Fraisse and Cockrem, white laying hens exhibited higher plasma corticosterone levels after handling compared to brown laying hens, peaking after 15 min (white hens 79.910 ng/mL; brown hens 53.702 ng/mL) following a handling stressor. The hens were picked up and held in an inverted position (baseline corticosterone at 0 min in both groups < 2.7 ng/mL) during the study [28].

Similarly Broom recorded an increased corticosteroid level in the plasma of hens that were taken from a cage and placed in an inverted position with their heads down, indicative of a stress reaction [108]. In contrast, the corticosteroid concentration of hens that were taken upright from the cage remained constant. Delanglez et al. recorded various parameters from 3000 individual laying hens obtained from seven different commercial cage-free flocks, which were either caught and carried upright or caught by one leg and carried inversely by the legs [109]. The interaction between the hen and the catcher when catching and

carrying by the legs on the 7-point Likert scale was rated significantly better ('calm and efficient') compared to upright catching ('calm and less efficient' $p < 0.001$) [109]. However, the upright catching and carrying took almost twice as long (8.17 h/1000 hens/person) compared to catching and carrying by the legs (4.75 h/1000 hens/person; $p = 0.011$). In the slaughterhouse, the health status of the hens was then examined for several animal-based parameters (bruises on wings, wing tips, breast, legs, fractures of wings and legs, dead on arrival). The only statistically significant difference was found in the prevalence of bruises on the wings, where 0.6% of the hens caught upright had fewer bruises compared to the inversely carried hens ($1.13 \pm 0.63\%$ vs. $1.73 \pm 0.70\%$) [109]. It should be noted that although in this study all inversely positioned 94-week-old laying hens were only caught by *one* leg, there was no statistically significant difference in the prevalence of leg and wing fractures between the methods (0.01% and 0.06% for inverted and upright handling, respectively). This stands in direct contrast to studies from the 1990s, which reported that 14–29% of laying hens had bone fractures at the slaughterhouse [108,110,111]. Possible reasons for this could be that many of the earlier studies were conducted on hens from battery cages, which often showed osteoporosis due to their restricted movement [112]. More recent studies refer to hens that are kept in aviary systems, or at least in cage-free floor systems, so that they perform bone-strengthening movements daily due to the horizontal and vertical structural opportunities. Furthermore, genetic loci for osteoporosis could be identified and both breeding and nutritional measures have significantly improved bone stability [113–115]. The role of nutritional factors, age and genetic aspects in laying hens during egg production, in relation to bone fragility, have been demonstrated for decades. Absolute and relative nutrient deficiencies (most commonly calcium, phosphorus and vitamin D) as well as the usage of certain genetic strains and housing design can lead to inferior bone strength. Common syndromes such as Cage Layer Fatigue predispose hens to transport-related injuries including fractures [116–118]. Professional formulated and prepared diets should always focus on bone health. Especially when caring for hens beyond 100 weeks of age, providing macro- and micronutrients as required as well as the adequate physical form of the ingredients is crucial to avoid transportation damage. Additionally, earlier studies reported rough handling during the extraction of individual hens from the cage, which contributed significantly to bone fractures [119]. A closer analysis of the various catching and transport steps revealed that most fractures occurred during poultry unloading at the slaughterhouse, and less during the actual catching, loading and transport [120]. Hence, studies on the prevalence and causes of fractures at the slaughterhouse from 35 years ago no longer reflect the situation at the time of this review.

6. The Consequences of the Upright and Inverted Catching and Carrying Method for the Flock, the Employees, and the Business

6.1. Consequences for the Flock

Catching and carrying usually affects the physical and psychological well-being of the birds as well as their behavior in a negative manner [110,121,122]. Birds are generally stressed by unusual interactions with humans, which can manifest in an increased frequency of fecal droppings, elevated heart rates, and increased corticosteroid concentrations [30,33,121,123–127]. Even animals that are seemingly unharmed during transport will likely experience fear, exhaustion, thirst, hunger, and/or potentially trauma. Therefore, the well-being of all animals to be transported is directly negatively affected by the intensity and duration of capture and transportation stress itself and indirectly affected by the accompanying prolonged period of food and water deprivation [107]. The stress response of catching can overshadow the animal's reaction to food and water deprivation or the inverse or upright carrying process. Consequently, it is crucial that holistic flock

welfare parameters need to be included in the welfare assessment of birds handled in flocks [39,121,128]. Such flock parameters include, for example, transport-related deaths, with the incidence of transport-related deaths being selected by the EFSA as an “iceberg indicator.” A current study was, however, unable to find significant differences in the rate of death on arrivals (DOA) between the various catching methods (DOAs (%)) inverted 0.25 ± 0.06 , upright 0.14 ± 0.07 , mechanical 0.20 ± 0.09 , $p = 0.18$). The duration of the entire transport process was controlled and therefore comparable [3]. A previously commissioned expert group of the EU Commission also deemed the recording of signs of thermal stress as relevant [129]. The number of dead animals upon arrival at the slaughterhouse depends on various factors such as the fitness for transport, the genetic makeup, the number of medical pre-treatments of the flock, etc. The DOA is also directly related to the conditions of transport and the stressors that occur during this transport [130–132]. While under ideal conditions (temperature control, provision of feed and water), losses of less than 0.005% (day-old chicks) can be ensured even with transport times exceeding 24 h (personal communication with Chickliner), pullets, laying hens, and broiler chickens are usually transported without feed and water for hygiene reasons. The main factor for delayed physical recovery after transport, or death during transport, is therefore the total transport duration [133,134]. The stressful influences, as well as the overall transport time, should be kept as low and short as possible, which gives particular importance to the duration of the capturing and loading process [33,39,108,135].

The influence of the total transport duration is evident, for example, in the mortality rates of young hens, laying hens, and broilers, which are all predominantly caught and carried inversely by their legs in Europe. The incidence of poultry species arriving dead at the destination is, despite similar capture methods, generally highest for spent laying hens [136,137]. An Italian study analyzed 1.266 billion broilers, 118 million turkeys, and 54 million laying hens over a period of four years, and recorded that, on average, 0.35% of broilers, 0.38% of turkeys, and 1.22% of laying hens arrived dead at the slaughterhouse [136]. These figures are comparable to other European countries and raise the question of why spent laying hens die more frequently. For example, in Germany, 0.08–0.13% of broilers, 0.10–0.12% of turkeys, and 0.25–0.36% of laying hens were recorded as transport-related deaths between the years 2019 and 2022 [137]. Various factors need to be considered when interpreting the data: in the EU (and Italy), approximately 40% of all laying hens are currently kept in cages, and injuries when catching and removing caged hens grasping on often osteoporotic legs are substantially more common and severe than in hens caught in cage-free housing systems. Cage-free housing can exclusively be found in Germany, Denmark, Finland, France, Luxembourg, The Netherlands, Austria and Sweden inside the EU, as well as the UK, Norway and Switzerland outside the EU [112,138–140]. Compared to broilers and turkeys, the overall transport time for laying hens is generally longer due to poor slaughterhouse infrastructure (low number and therefore unfavorable geographical distribution). At the slaughterhouse, one may also expect extended waiting times due to the poorly calculable arrival times of long transport routes. This contributes to the possibility that the legal total transport duration of 12 h may be exceeded [7]. It is therefore particularly critical for laying hens to keep the catching time as short as possible in order to not extend the overall transport time further. Both catching upright and catching with two legs significantly prolong the transport time of a flock of laying hens compared to catching by one leg. Delanglez et al. recorded the required duration of 8.2 ± 3.2 person-hours/1000 upright caught and carried hens compared to 4.8 ± 2.0 person-hours/1000 hens caught by the feet and carried inversely ($p = 0.011$) [109]. Likewise, catching broiler flocks by two legs took twice as long as catching by one leg according to Wessel et al. and Langkabel et al. [70,106]. However, Kittelsen et al. compared the catching of broiler flocks by

two legs with catching in an upright position [107]. Overall, catching by two legs took significantly longer. When comparing the two methods in a differentiated way regarding slow-growing and conventionally growing broiler chickens, only in the case of conventionally raised broiler chickens did the upright catching lead to a significantly faster loading time. [107]. A recent study investigated the animal welfare and ergonomic and economic effects of employing different catching methods for broilers (inverted, upright, mechanical). The authors found that the upright catching method required more person-hours per 1000 broilers [3]. Specifically, the upright catching required an additional 0.63 h per 1000 broilers compared to the inverted catching method and 1.08 h per 1000 broilers compared to the mechanical catching of the animals. To fill the containers with the same number of broilers, an additional 2.91 min were needed for upright handling compared to inverted handling and additional 4.41 min compared to mechanical catching [3].

The two-legged catching and loading of poultry also resulted in significantly higher variability in stocking density in the transport containers (31.2–64.0%), whereas one-legged catching provided a more uniform loading of the transport containers (5.2%) [70,104]. This parameter, which is only recorded when a flock and not an individual animal is loaded, is also an indicator of indirect consequences of a catching method that must be considered for overall animal welfare. An uneven stocking density in the transport containers leads to inadequately high stocking densities of animals that suffer from immobility, heat, or cold stress [141]. The large variation in stocking density in the transport containers can be explained by the fact that catching by two legs tires the catching team, and therefore the employees cannot perform their tasks with the usual and necessary concentration, caution and due diligence [106,107,142–144]. Animal welfare during the catching, carrying and transportation of a flock is influenced by multiple factors. The care taken by the catcher and transport duration are the two most relevant factors [131,133,139,142]. However, when these variables are specifically controlled for and only the effect of the catching method is examined, it becomes evident that the number of deceased hens is comparable between the catching methods (one-legged vs. upright) (0.23% vs. 0.22%). The deceased hens succumbed not to traumatic catching injuries but rather to unrelated underlying diseases [109]. Similarly, the different catching and carrying methods for broilers did not show any influence on the incidence of animals with bruises or on the mortality rate (0.13%), whereas the loading time contributed significantly ($p = 0.03$) to this [135,145].

Studies on injuries and fatalities during the loading of pullet flocks, which are typically caught by their legs and carried invertedly, are scarce. They regularly arrive at their destination in good condition, begin to eat on the same day, experience only marginal weight loss, and soon start laying eggs ([146,147]; personal observations). This may be because pullets, due to their dense plumage, can maintain their homeostasis relatively well during transport and generally appear to be relatively resilient to handling stress [132,147]. Pullets are often raised near their laying houses, resulting in shorter transport times compared to spent laying hens, who need to travel to processing plants. It is also important to note that pullets are more valuable than spent hens, so they are likely caught and loaded more carefully [132]. Subsequently, the inverse catching method itself may not necessarily result in diminished animal welfare. Rather, the care of handling as well as the logistics related to the poultry transport is critical to ensure the best possible welfare of the transported birds. A summary of the key studies on layer and broiler catching, which are cited in Sections 5 and 6, is provided in Table 2.

Table 2. Summary table of the key studies on layer and broiler catching.

Reference	Type of Bird	Body Weight/Age	Catching Method	Sample Size (Flocks/Birds)	Welfare Indicators	Main Findings
Delanglez et al. (2024) [109]	Laying hens	94 weeks	One-legged (inverted) and upright	7 flocks: ~3000 hens per method and flock	Behavior, noise, wing flapping, catcher–bird–interaction, DOA, injuries	Upright method took almost twice as long; 1.8 times more expensive; lower prevalence of wing flapping and wing bruises
Gerpe et al. (2021) [100]	Laying hens	72–85 weeks	Upright method (for baseline assessment); various catching methods combined with inverted carrying	15 flocks, 603 hens in total	Active and passive collisions, injuries (bone and muscle)	8.1% of the hens showed severe injuries (fractures, muscle damage, increased corticosterone) due to collisions when carried out by non-professional catchers
Delanglez et al. (2025) [3]	Broilers	~42 days	Inverted, mechanical and upright	15 flocks, 5000 broilers per method and flock	Behavior, noise, wing flapping, catcher–bird–interaction, DOA, catch damage	Upright method required 0.63 additional person-hours per 1000 broilers, 1.5 times more expensive, less wing flapping, better catcher–bird interaction
Kannan et al. (1996) [39]	Broilers (male)	42–49 days	Inverted and upright	Research facility with 280 broilers	Plasma corticosteroid concentrations	Upright handling led to significantly lower corticosteroid concentrations; crating stress can mask the prior handling stress

Table 2. Cont.

Reference	Type of Bird	Body Weight/Age	Catching Method	Sample Size (Flocks/Birds)	Welfare Indicators	Main Findings
Kittelsen et al. (2018) [107]	Broilers	1.3–1.5 kg 33–44 days	Two-legged (inverted, up to 3 per hand) and upright (2 per catch)	2 flocks (different strains): 3951 broilers	Fractures, DOA, stocking density	Catching by two legs took significantly longer, broilers were more restless, showed more wing fractures; less consistent crating density
Langkabel et al. (2015) [106]	Broilers	1.9–2.5 kg 35–42 days	One-legged (inverted) and two-legged (inverted)	Farm 1: 38,500 broilers Farm 2: 29,500 broilers	Lesions (hemorrhages, wing fractures), wing flapping	Catching by two legs took longer in general; no effect on the prevalence of bruises or fractures in the carcass
Wessel et al. (2022) [70]	Broilers	2.3–3.1 kg 30–40 days	One-legged (inverted) and two-legged (inverted)	12 loadings, average of 8679 broilers per loading	Behavior, injuries	Catching by two legs took twice as long; significantly higher variability in stocking density in the transport containers (two-legged method); one-legged method led to higher chances for epiphysiolysis

DOA = Death on arrival (at the slaughterhouse).

6.2. Consequences for Employees

Already in the 1980s, the impact of poultry capturing on employees was documented [1]. Kettlewell and Turner calculated that poultry catchers carried 5–10 tons of load in a work shift when lifting up to 1500 individual birds per hour during a 5 h shift [1,148]. Manual capturing is a tiring, monotonous, and dusty activity [37]. Millman et al. conducted interviews with catchers and concluded that manual poultry catching is one of the most challenging jobs in poultry farming [149]. Employee-friendly capturing is therefore particularly relevant to the health of employees. The activity is usually carried out at night and in the dark to support the bird's welfare, which, however, poses an additional burden for employees [150]. The two-legged catching of broilers was perceived by the employees as more tiring than the one-legged catching, because more time had to be spent in a squatting position [106]. Other broiler catchers also reported that it was more cumbersome and therefore took longer to sort the two legs of the animals. If given the choice between two-legged and upright catching, the catchers preferred to lift the broilers individually by their bodies and carry them upright, even though the more frequent bending was described as tiring [107]. However, even with upright catching, a significantly increased restlessness of the consistently upright caught broilers was observed starting from the 5th hour of work. This was likely caused by tired catchers and, accordingly, more catching attempts during the grasping [2]. This is in line with a recent study from 2025, which quantified the physical demand of manual poultry catching ergonomically [3]. Both the upright and the inverted methods received very poor scores. The upright catching method performed slightly better from an ergonomic perspective (total average score 22 inverted vs. 19 upright, 0–11 represents low risk, 12–21 medium risk, >22 high risk) if performed for the same number of times (a total of nine repetitive catching events) [3]. However, the catchers themselves described the upright catching method as more exhausting and painful. Additionally, during a catcher survey for laying hens, the employees rated the upright catching method as difficult (53%) to very difficult (27%) to learn. Especially with laying hens caught from aviary systems, which require less bending than catching the broilers from the ground, the majority of catchers prefer catching one leg over upright catching [109].

6.3. Consequences for Operations

Recent calculations show that approximately twice as many person-hours are required per 1000 hens for upright capture compared to single-leg inverted capture if it should be conducted within the same time [3,70,109]. That lead to an average increase in labor costs of 180% for the upright method (€369.4 per 1000 upright laying hens vs. €206.5 per 1000 inverted laying hens) and an average increase of 160% in forklift costs (€27.8 per 1000 upright laying hens vs. €17.5 per 1000 inverted laying hens) and truck loading costs (€29.8 per 1000 upright laying hens vs. €18.8 per 1000 inverted laying hens) due to the additional operating time. Hence, the total cost for upright capture (€427/1000 laying hens) was 1.8 times more expensive than for inverted capture (€242.8/1000 laying hens). That would result in an average additional cost of €0.19 per laying hen caught upright and an additional price increase of €0.0005 per egg [109]. Regarding broiler depopulation, a study conducted in Belgium modeled the average labor costs to be 1.5 times higher for the upright method than the inverted method (€67.2 per 1000 upright broilers vs. €43.4 per 1000 inverted broilers). Forklift (€12.5 per 1000 upright broilers vs. €8.7 per 1000 inverted broilers) and truck loading costs (€14.3 per 1000 upright broilers vs. €10 per 1000 inverted broilers) were on average 1.4 times higher when using the upright method due to additional operating time. The average total costs for depopulating 20,000 broilers were 1.5 times higher for the upright method (€1880 upright vs. €1242 inverted), resulting

in an average extra cost of €0.032 per broiler and a price increase of €0.012 per kg live body weight when applying the upright method [3]. To ensure that the extended capture time does not negatively impact the well-being of the animals, at least twice as many employees would need to be deployed per shift, with the corresponding financial consequences. In addition, the sheds would need to allow for the additional traffic of moving more crates and people at the same time. Furthermore, it was suggested that additional staff would be needed who are solely responsible for counting the loaded animals and organizing the logistics. Thus ensuring that, despite rapid signs of fatigue, a uniform stocking density in the transport containers is maintained (see Section 6.1) [107,141]. Experienced and trained staff significantly reduces the injury rate of the animals to be transported and is essential for successful poultry transport [151,152]. Therefore, it is of great interest for operations to employ not only sufficient but also suitable, trained and experienced staff for catching and loading poultry. The availability of suitable personnel is deemed critical; the attractiveness of poultry catching would likely be further reduced by two-legged/upright catching of the animals due to the survey results described in Section 6.2. Within a study from 2024, farm managers from various poultry farms (both layers and broilers) indicated that they preferred the inverse catching method over upright catching [98]. If the catching duration is not compromised, the gentlest method for transferring poultry flocks is one that requires the least interaction with humans [53]. In the case of broilers, this would be the uneventful application of carefully conducted mechanical catching methods, and in the case of turkeys, it would be driving the animals directly into the loading unit [89,103]. These methods are already being used in countries wherever shed construction and logistics allow. For laying hens, there is currently no mechanical alternative in many countries. This is due to cage-free housing in sheds with multiple aviary levels and build-in furniture such as nest boxes, feeders and drinker lines.

In summary, there are several, sometimes contradictory, recommendations from public bodies and consortia on how the manual catching and carrying of birds should be conducted. The large variety of birds' behavior, skillset of the catchers, housing systems and logistical infrastructure require flexible solutions for individual circumstances. Objective data collection of suitable and holistic animal welfare parameters during various steps of the poultry transport is complex and must be interpreted in a differentiated manner in order to allow for transferable conclusions and prove causal relationships. Poultry transport consists of many consecutive individual actions, all of which influence the occurrence of suffering, pain, and physical damage to varying degrees. Integrated and holistic measurements on multiple levels, considering the species, duration, and severity, are preferred for capturing the actual condition of poultry. The corticosteroid concentration, indicative of the stress response, is lower in broilers and laying hens that are caught and carried upright than in inversely carried birds. These values are often recorded in scientific institutions with small test groups/cage systems. Under these husbandry conditions, upright catching and carrying represent the least stressful method [153]. Due to the significantly larger floor spaces, possibly with vertical aviary structures, these results are not transferable to the stress level of the catching process in larger flocks. Wing flapping accompanied by bruises is by far the most common injury during the manual catching and transporting of broilers and laying hens. Several studies conclude that the significance of the catching method for animal welfare is secondary for both broiler flocks and non-caged laying hen flocks, as long as injuries from wing flapping (through hand placement or carrying multiple animals) as well as from passive and active collisions (through careful and swift catching) are avoided. Birds exhibit various anatomical and physiological features (air sacs with variable volume, rigid lung attached dorsal within the coeliac cavity, no muscular diaphragm) that enable their respiration in different positions. Restricted movements of the chest can lead to the

death of the animal. When birds are in a supine position, physiological parameters such as tidal volume change, but not necessarily the minute volume of respiration or the oxygen saturation in the blood, which is why this position is regularly applied during treatments under anesthesia. Nevertheless, these findings cannot be extrapolated to the inverted position. To the best of our knowledge, there are currently no studies on the anatomical, physiological and clinical consequences of birds placed in an inverted position. The decisive factor for animal injuries is the exhaustion level of the trained and careful catchers, which directly correlates with the duration of the catch and thus the catching efficiency. Therefore, the method of catching should be chosen for the welfare of the animals and people in such a way that the total catching duration of the flock is as careful and short as possible under the given circumstances. This plays a particular role in laying hen flocks. Due to poor slaughterhouse infrastructure, the overall transport times for spent laying hens are generally longer in Europe. Therefore, catching birds at the first attempt and gentle, rapid loading is essential. For laying hens, upright catching takes significantly longer compared to catching by the legs. In broilers, catching by two legs or upright takes up to twice as long compared to catching by one leg. Catching in an upright position (in laying hens) or catching by two legs (in broilers) has a significantly adverse effect on workplace safety and the job satisfaction of the catchers. Poultry operations must anticipate increased costs and a doubled necessity for qualified and capable staff when employing upright catching (laying hens) or catching by two legs (broilers).

7. Conclusions

Due to the various influences that affect the animals during poultry transport, as well as the absence of a 'best' catching method, it would be advantageous for the poultry to carry out a consistent, transparent, and traceable central data collection on poultry health and welfare at several critical control points including the destination. This should include specific welfare and health indicators for each step of the transport process. Implementation would have an immediate and direct impact on animal welfare and enable all parties in the transport chain (including farmers, catching teams, transport companies, slaughterhouses, and the responsible authorities) to effectively design the poultry transport in such a way that animal welfare is maximized while ensuring occupational safety and employee satisfaction. Further studies focusing on the anatomical and physiological effects in different poultry species during short-term inverted positioning including information on anatomical, physiological and behavioral indicators are highly warranted.

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Abbreviations

The following abbreviations are used in this manuscript:

EU	European Union
EFSA	European Food Safety Authority
ABM	animal-based measure
e.g.,	exempli gratia (for example)
DOA	Death on Arrival

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